

Thilina H. Weerakkody, Ph.D.

Robotics, Dynamics & Control Engineer

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Professional Summary

Robotics and control systems engineer with a Ph.D. in Mechanical Engineering and research experience spanning non-linear dynamics, advanced control, robotic systems, artificial muscles, embedded systems, and intelligent automation. Experienced in developing physics-based dynamic models, robust and adaptive controllers, robotic mechanisms, simulation frameworks, and real-time experimental platforms. Background includes soft and bioinspired robotics, assistive and medical robotic systems, underwater robotics, high-throughput robotic automation, and physics-informed machine learning. Proficient in MATLAB/Simulink, Python, C/C++, ROS 2, and physics-based robotic simulation.

Core Expertise

Robotics & Autonomous Systems: robot kinematics and dynamics, multibody systems, trajectory generation, motion control, ROS/ROS 2, robotic simulation, motion planning, soft and bioinspired robotics

Dynamics & Control: nonlinear dynamics, classical and modern control, robust control, adaptive control, model predictive control, sampled-data control, system identification, estimation, stability analysis

Physics-Based & Intelligent Modeling: physics-based modeling, physics-informed machine learning, parameter estimation, system identification, data-driven modeling, simulation and optimization

Embedded & Experimental Systems: real-time control, embedded C/C++, microcontrollers, sensors and actuators, hardware integration, instrumentation, experimental validation and rapid prototyping

Technical Skills

Programming: MATLAB, Python, C/C++

Robotics: ROS 1/ROS 2, Gazebo, RViz, MoveIt, MuJoCo, PyBullet, URDF/Xacro

Modeling & Control: MATLAB/Simulink, nonlinear dynamics, multibody dynamics, robust/adaptive control, MPC, system identification, estimation/filtering

Machine Learning & Vision: PyTorch, TensorFlow, scikit-learn, NumPy/SciPy, OpenCV, physics-informed ML

Embedded Systems: Arduino, Raspberry Pi, Teensy, embedded C/C++, SPI, I²C, UART, motor drivers, real-time control

Engineering Tools: SolidWorks, Fusion 360, COMSOL, ABAQUS, LabVIEW, Git/GitHub, Linux, Docker

Sensors & Prototyping: encoders, load cells, thermocouples, EMG, cameras, optical motion capture, 3D printing, molding, soldering and rapid prototyping

Professional Experience

Senior Lecturer (Grade II), Mechanical Engineering

2026–Present

University of Moratuwa, Sri Lanka

- Conduct teaching and research in control systems, robotics, nonlinear dynamics, mechatronics, and intelligent systems.
- Develop research programs in nonlinear and advanced control, autonomous and robotic systems, multibody dynamics, and physics-informed machine learning.
- Supervise research projects involving robotic manipulators, cable-driven robots, compliant mechanisms, autonomous systems, embedded control, and simulation-based control design.
- Develop MATLAB/Simulink, Simscape Multibody, and ROS-based experimental and simulation frameworks for robotics and control education and research.

Postdoctoral Scholar

2024–2026

California NanoSystems Institute (CNSI), University of California, Los Angeles

- Contributed to the development and optimization of high-throughput robotic automation platforms for radiochemistry and biomedical applications.
- Developed modeling and analysis workflows for coupled thermal, optical, mechanical, and measurement processes affecting robotic platform performance.
- Developed Python and MATLAB tools for calibration, parameter estimation, experimental data analysis, system characterization, and performance optimization.
- Supported integration and experimental validation of automated robotic subsystems for high-throughput operation.

Ph.D. Research Assistant – Controls, Robotics and Dynamics

2019–2024

University of Iowa

- Developed nonlinear physics-based dynamic models for artificial-muscle-driven robotic systems and validated models through experimental system identification.

- Designed robust, adaptive, and sampled-data control algorithms for nonlinear actuators and robotic systems under uncertainty, disturbances, hysteresis, and thermal effects.
- Developed robotic systems for assistive, rehabilitation, underwater, soft robotic, and bioinspired applications.
- Built MATLAB/Simulink and Python simulation frameworks for nonlinear dynamics, controller design, parameter estimation, transient-response analysis, and experimental validation.
- Implemented real-time control algorithms using embedded microcontroller platforms and integrated sensors, actuators, data-acquisition systems, and experimental instrumentation.

Research Engineer – Medical Robotics

2020–2023

Juggernaut Life Sciences / University of Iowa Collaboration

- Designed robotic and electromechanical mechanisms for automated biomedical sample-preparation systems.
- Developed CAD models, mechanical prototypes, sensing architectures, and control concepts for automated medical devices.
- Contributed to patented technology for semi-automated immunolabeling and sample preparation.

Research Assistant – Bionics & BioRobotics

2016–2018

University of Moratuwa, Sri Lanka

- Designed and prototyped lower-limb prosthetic and adaptive foot mechanisms using kinematic, dynamic, embedded-system, and rapid-prototyping methods.
- Integrated sensors and actuators with embedded controllers and implemented real-time data acquisition, filtering, and control logic.

Education

Ph.D., Mechanical Engineering, University of Iowa

2019–2024

Dissertation: Design and Control of Artificial Muscles for Robotic Applications

Focus: nonlinear dynamics, physics-based modeling, robust/adaptive control, system identification, and robotic systems.

B.Sc. Engineering (Hons.), Mechanical Engineering, University of Moratuwa, Sri Lanka

2011–2016

Thesis: Development and Control of a Modular Artificial Shoulder Prosthetic Device

Diploma in Information Technology (HEQ), British Computer Society (BCS), UK

2012–2014

Selected Research, Publications & Patent

- **T. H. Weerakkody** et al., “Robust Control of Shape Memory Alloys for Assistive Robotics Applications,” *Journal of Shape Memory and Superelasticity*, 2025.
- R. Mamman, **T. H. Weerakkody**, and C. Lamuta, “Bioinspired Active Vortex Generators for Underwater Flow Control,” *Robotics Reports*, 2025.
- P. Kotak, S. Maxson, **T. H. Weerakkody**, V. Cichella, and C. Lamuta, “Octopus-Inspired Muscular Hydrostats Powered by Twisted and Coiled Artificial Muscles,” *Soft Robotics*, 2023.
- **T. H. Weerakkody** et al., “Modeling and Control of Twisted and Coiled Artificial Muscles for Soft Robotics,” *Meccanica*, 58(4):643–658, 2023.
- C. Greco, **T. H. Weerakkody** et al., “Lightweight Bioinspired Exoskeleton for Wrist Rehabilitation Powered by Twisted and Coiled Artificial Muscles,” *Robotics*, 12(1):27, 2023. **Editor’s Choice Article.**
- M. Hammond, V. Cichella, **T. H. Weerakkody**, and C. Lamuta, “Robust and Adaptive Sampled-Data Control of Twisted and Coiled Artificial Muscles,” *IEEE Control Systems Letters*, 6:1232–1237, 2021.
- M. T. Rahman et al., “Semi-Automated Immunolabeling Systems and Associated Devices and Methods,” U.S. Patent Application [US 2025/0231210 A1](#), 2025.

Full publication list: [Google Scholar](#)